Module 5: Estimation of Synergies and other M&A benefits

Outline

- 1. What are "synergies"?
- 2. What are other possible M&A benefits?
- 3. Two methods to estimate synergies using DCF

Recall what we have discussed

For regular investors → Standalone Value

For acquirers: → Standalone Value + Synergies + Other M&A benefits

$\underbrace{\text{Synergies} + \text{Other M\&A benefits}}$

“Merger Gains”

Why acquirers are willing to pay a premium

Bill Gates' Own Words on the Microsoft-Linkedin Deal in 2016 https://www.youtube.com/watch?v=LSndZFVGkfg

- "(Linkedin team) so **combined** with what Microsoft does, working with people in productivity helping professionals communicate until be a lot of **synergies** there…"
- It seems that synergy is an important factor in the Microsoft-Linkedin merger.
- M&As are supposed to create **synergy**.
- But what is **synergy**?

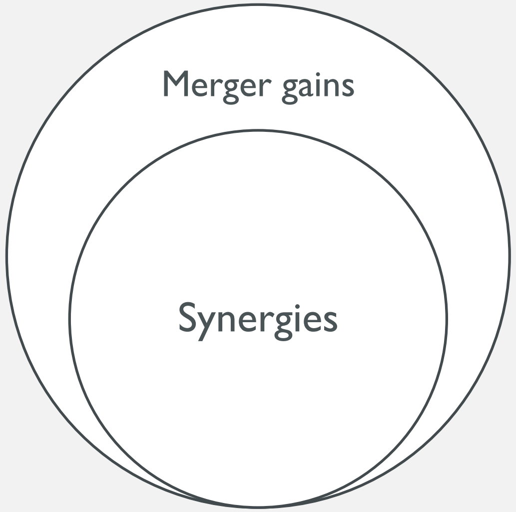
What is “synergy”?

- “1+1 >2”
- Synergy is the *__increase__* in value generated by the __combination__ or __integration__ of two companies’ operations.
- Synergies are the most frequently cited motive for strategic acquirers to pursue M&A deals.
- Synergies might be from different sources. However, we should be *careful* with the claim of synergies because
  - 1. The valuation of synergies requires a lot of guesswork.
  - 2. Synergies are difficult or will take a long period to materialise.
  - 3. Some typess of synergies could be difficult to quantify.
  - 4. Acquirers’ top management tends to be overoptimistic about synergies.

Example of other M&A benefits (than synergies)

- *<u>Synergy</u>* refers to the value enhancement that can be achieved when the acquirer and the target' are integrated or combined.

- Note that the acquirer can improve the target's value even in the <u>absence of integration</u> by
  - removing inefficient management of the target
  - selling idling or underutilized assets of the target
  - improving the target's operating efficiency
  - increasing the debt level of the target if the target company is underlevered
  - …….
- <u>Merger gains</u>: synergies and other M&A benefits.
  - Note "synergies" and "merger gains" are often used interchangeably in practice.



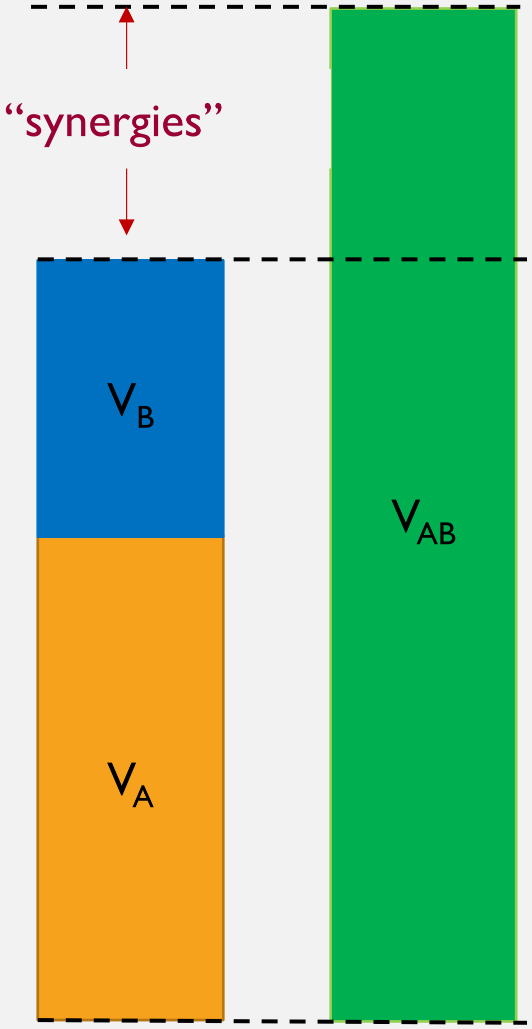
The Conceptual Framework of Valuation of Synergies (and merger gains)

- $$Synergies = V_{AB} - (V_A + V_B)$$

The target's
standalone total firm value

The acquirer's
standalone total firm value

Combined firm value
including synergies



How can we estimate the V_{AB} using the <u>DCF</u> approach?

- **Step 1: Identifies the specific sources of synergies**
- Step 2: Quantify and include the synergies in the DCF valuation process (e.g., FCFFs, WACC, g, etc.)
- Step 3: Use DCF to estimate the value of the combined company's **total assets** that **includes synergies**.

Live Case Series: #2

- Following live case series #1 (note that you can change acquirer and target companies if necessary):

1. Identify potential specific synergy sources between the proposed acquirer and target.
2. Assess the likelihood of materializing each synergy and highlight key challenges.

Specific synergies in M&As?

Synergies

- Lower material cost
- Lower labour cost
- Improved supply chain efficiency
- Shared overhead costs
- Exclusive/cheaper access to patents/R&D
- Geographic expansion
- Greater pricing power
- Cross-selling

Operating synergies

- Lower cost of debt
- Greater debt capacity
- Reduced taxes
- Lower cost of equity
- Better access of equity capital

Financial synergies

Categories of synergies

- Operating Synergies (revenue/costs):
  - (a) Costing Saving,also called *"Hard Synergies", e.g.:*
    - greater pricing power over suppliers
    - enhanced supply chain efficiency
    - cost sharing
    - the acquirer has efficient production lines and the target has strong R&D capability
  - (b) Revenue Enhancement, also called *"Soft Synergies", e.g.:*
    - greater pricing power over customers
    - the acquirer has strong marketing channels and the target has great products

- Financial Synergies (cost of capital/taxes):
  - (a) Tax saving: the acquirer is profitable and pays high taxes; the target is unprofitable and has tax losses for several years.
  - (b) Greater debt capacity: the combined firm can have a higher debt ratio due to reduced risk and stabler cash flows.
  - (c) Lower cost of debt/equity: the combined firm can have a lower interest rate/cost of equity due to reduced risk and stabler cash flows.
  - (d) Better access to equity capital: the combined firm can raise equity capital more easily or at a lower cost of equity.

How can we estimate the V_{AB} using the <u>DCF</u> approach?

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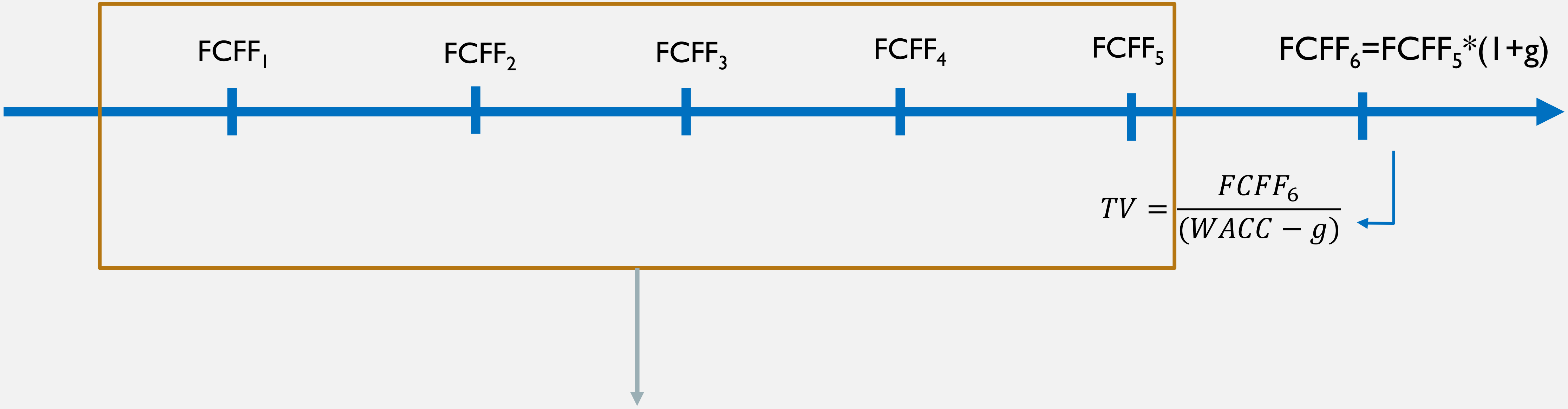
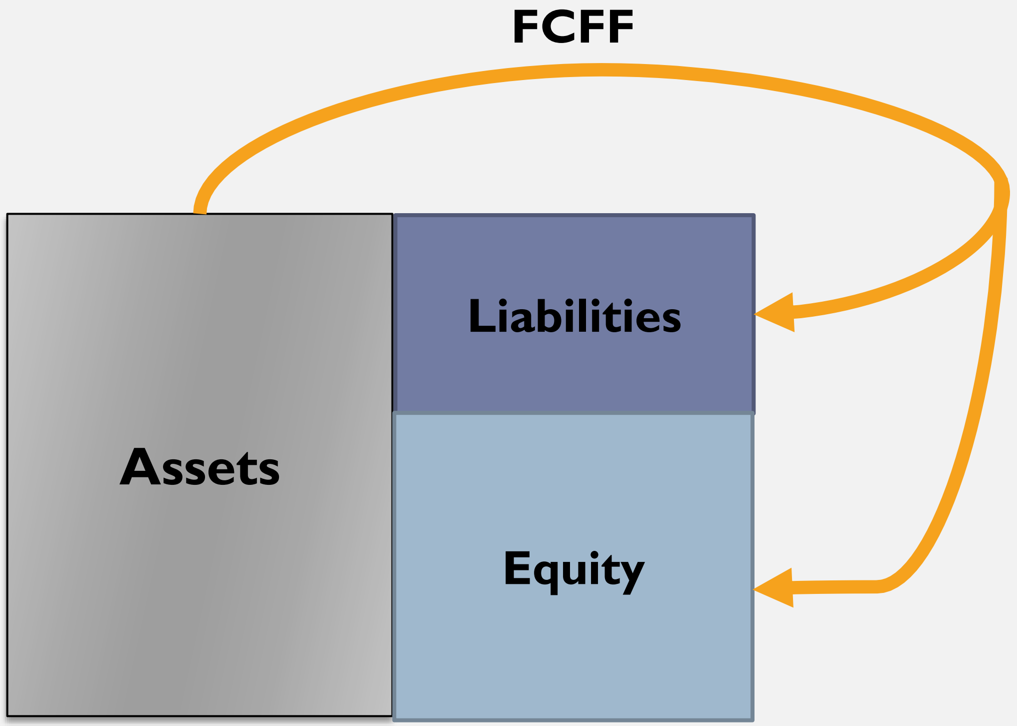
Recall how we valued a company's assets (or total firm value)

- Suppose we can forecast individual FCFFs to 5 periods from today. After that we assume the dividends will grow at a constant rate, g.



The value of the target's assets = PV of near term FCFFs + PV of Terminal Value

Recall How we estimate FCFFs



- $FCFF = EBIT(1 - T_C) + D\&A - \Delta NWC - CapEx$
- $FCFF = \boxed{EBIT(1 - T_C)} - \boxed{(CapEx - D\&A + \Delta NWC)}$

$"Net\ Reinvestment"$

$\boxed{Revenue\ - COGS\ - Operating\ Expenses\ - Taxes}$

Incorporate different synergies in the FCFFs

Lower material cost

Lower labour cost

Shared overhead cost

Exclusive/cheaper access to patents/R&D

Improved supply chain efficiency

FCFF =

Revenue

– COGS

– Operating Expenses

– Taxes

+ D&A

– ΔNWC

– Capital Expenditure

Geographic expansion

Greater pricing power

Cross-selling

Reduced taxes

Greater debt capacity

Better access of equity capital

Recall how we estimate long-term FCFF growth rate if we choose not to use long-term inflation or GDP growth

- $Long - Term\ EBIT\ Growth\ Rate = \boxed{Net\ Reinvestment\ Rate} \times \boxed{After\text{-}Tax\ Return\ on\ Capital}$

$$= \frac{CapExp\ - D\&A + \ \Delta NWC}{EBIT(1 - T_C)}$$

$$= \frac{EBIT(1 - T_C)}{BV\ of\ D + BV\ of\ E}$$

Incorporate synergies in the long-term EBITDA growth rate

- $Long - Term\ EBITDA\ Growth\ Rate = \boxed{Net\ Reinvestment\ Rate} \times \boxed{After\text{-}Tax\ Return\ on\ Capital}$

Greater debt capacity

Better access of equity capital

$$= \frac{CapExp\ - D\&A + \Delta NWC}{EBIT(1 - T_C)}$$

$$= \frac{EBIT(1 - T_C)}{D + E}$$

Recall how we find the discount rate

- Thus, discount rate should represent opportunity costs of all investors (i.e., debt (*D*) and equity (*E*)) holding claims against firm's assets
- Weighted average cost of capital (WACC)

Cost of equity

Pre-tax cost of debt

$$WACC = \frac{E}{D+E} \times \underbrace{r_e}_{\text{Cost of equity}} + \frac{D}{D+E} \times \underbrace{r_d}_{\text{Pre-tax cost of debt}} \times (1 - T_C)$$

Incorporate synergies in the WACC

Better access of equity capital

Greater debt capacity

$$WACC = \frac{E}{D+E} \times r_e + \frac{D}{D+E} \times r_d \times (1 - T_C)$$

Lower cost of equity/debt

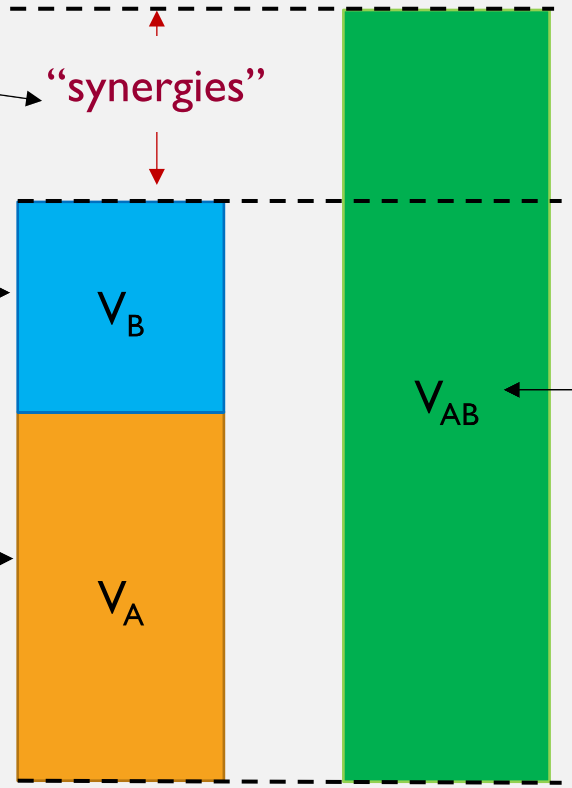
The valuation of synergies (and merger gains)

Method 1: The Long Way

Step 4: Synergies $=V_{AB} - V_A - V_B$

Step 1: The target's
Standalone total firm value
(Discount rate: use $WACC_B$)

Step 2: The acquirer's
Standalone total firm value
(Discount rate: use $WACC_A$)



Step 3: Combined total firm value including *synergies*
(Discount rate: use the combined firm's WACC)

Model Setup
(Assume that the acquirer and the target have zero non-operating assets)

Model Setup				
Categories	**Item**	**Acquirer**	**Target**	**Combined (No synergies)**
Firm Information	Equity Beta	1	1	1
	Cost of Debt (R_D)	5%	5%	5%
	Tax Rate	30%	30%	30%
	Weight of Debt (W_D)	10%	10%	10%
Near-Term FCFF Assumptions (Yr1 - Yr5)	EBIT @ t = 1	50	25	75
	EBIT Growth Rate	7.35%	7.35%	7.35%
	Net Reinvestment Rate	70%	70%	70%
Perpetual FCFF Assumptions (after Yr5)	Long-term EBIT Growth Rate	4.840%	4.840%	4.840%
	Net Reinvestment Rate	57.3%	57.3%	57.3%
Market Information	Risk-free Rate	2.00%		
	Market-risk Premium	7%		

Use the Spreadsheet " Model 5 synergy method 1 example blank"

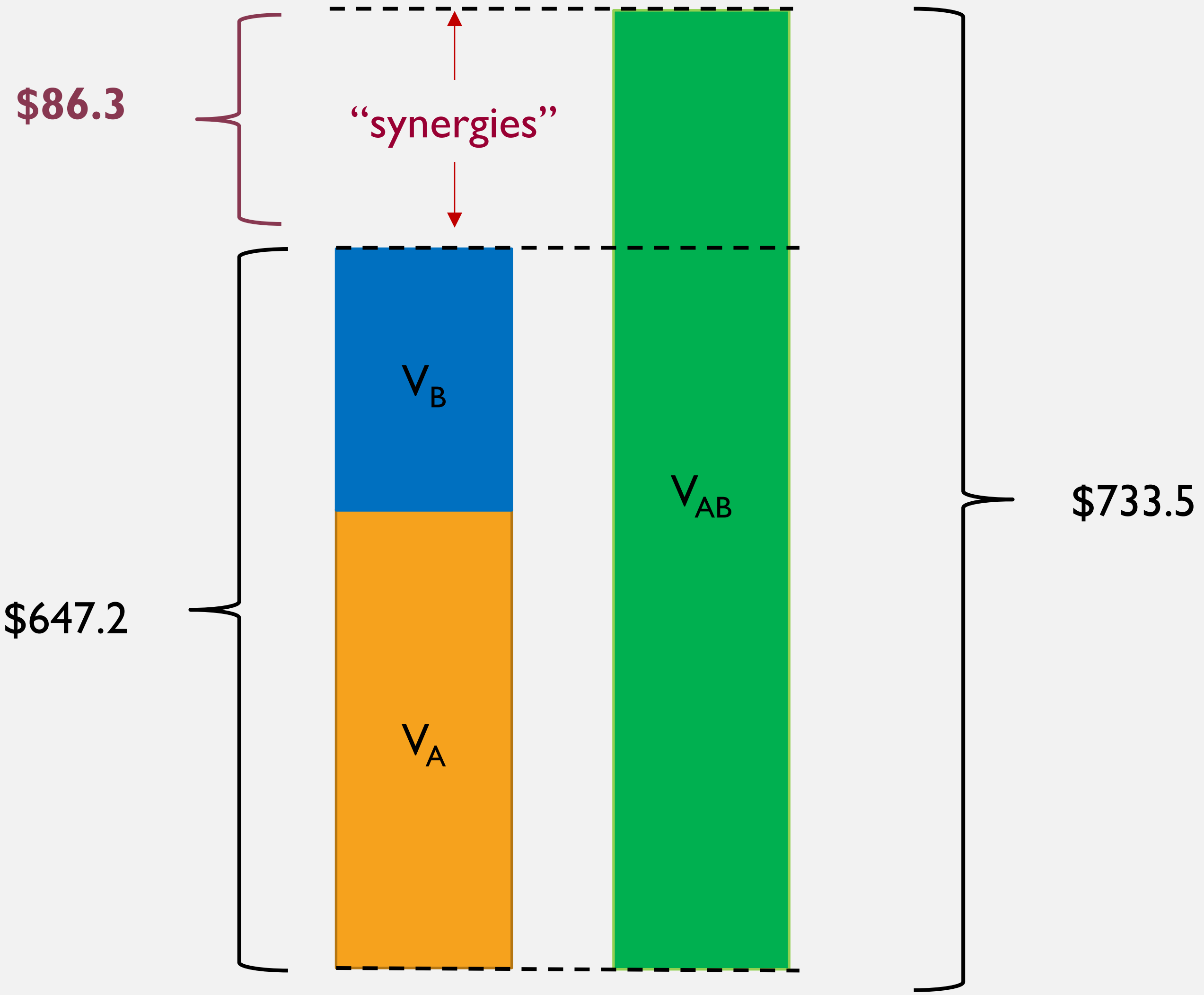
- Tasks:

- 1. Estimate the value of the "cost savings synergy" or hard synergy.

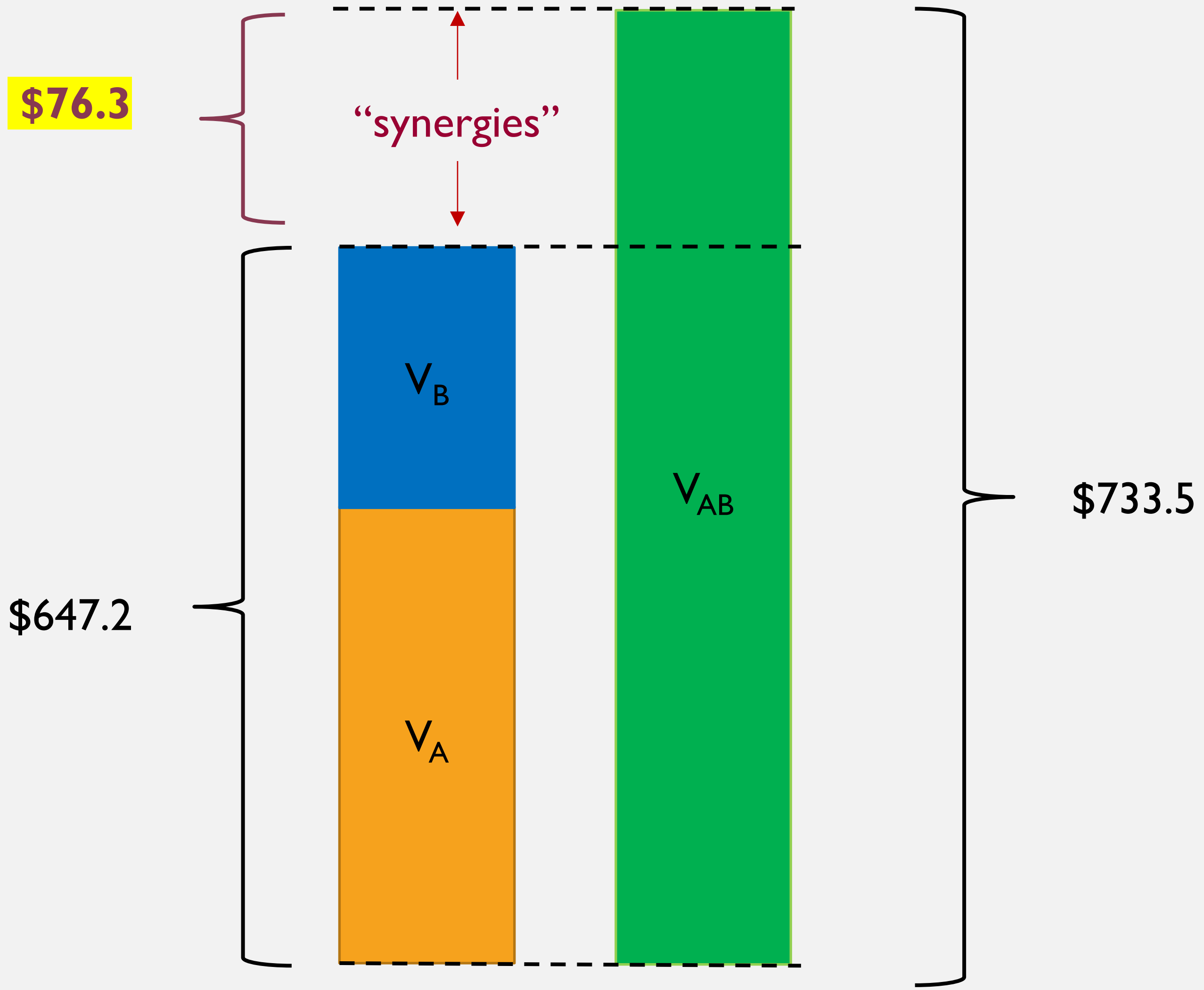
- 2. Estimate the value of the "revenue growth synergy" or soft synergy.

- 3. Estimate the value of the "cost of debt synergy" or financial synergy.

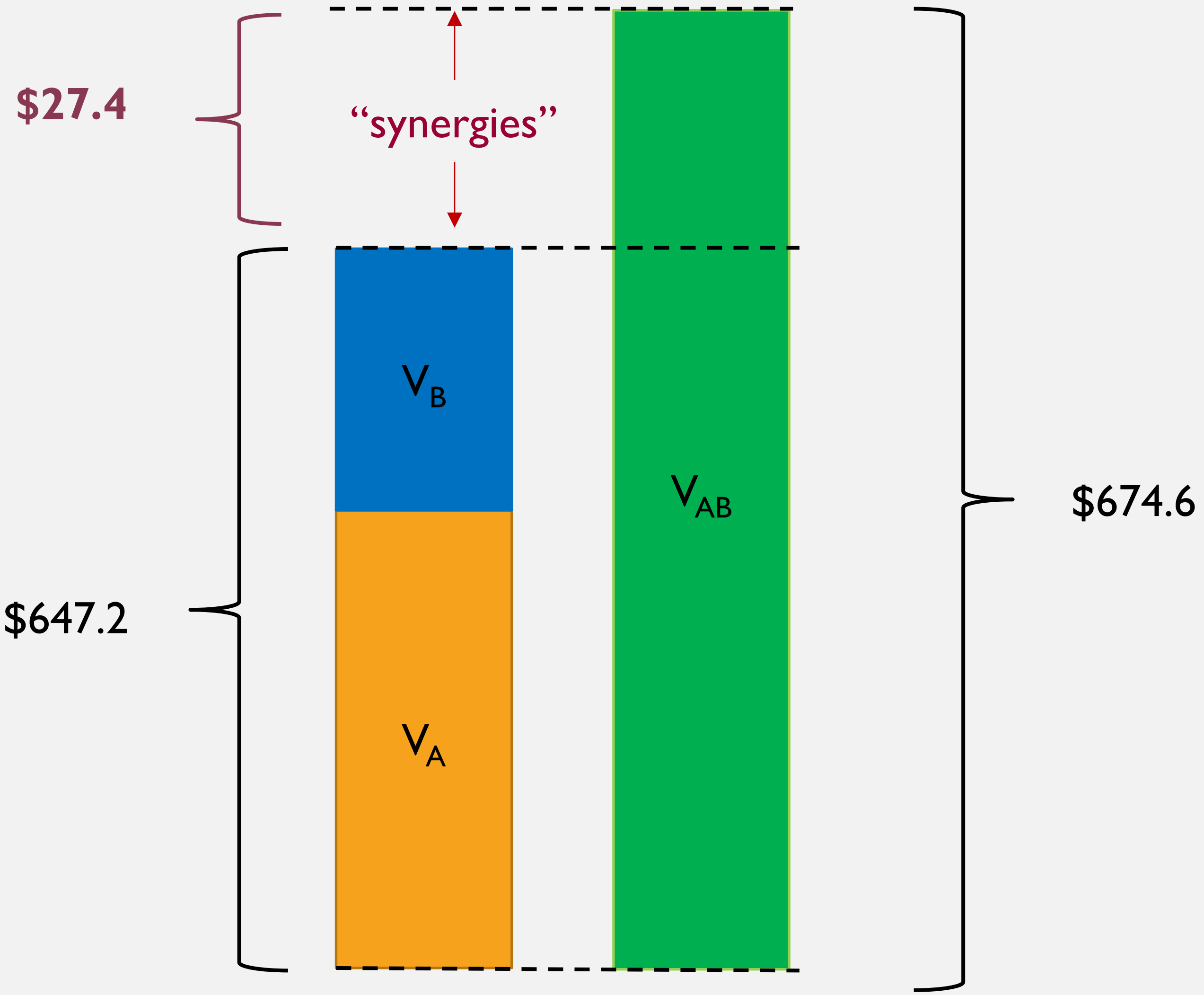
Cost Saving Synergy ("Hard Synergy") Modeling
(please refer to the spreadsheet)



Revenue Enhancement Synergy ("Soft Synergy") Modeling
(please refer to the spreadsheet)

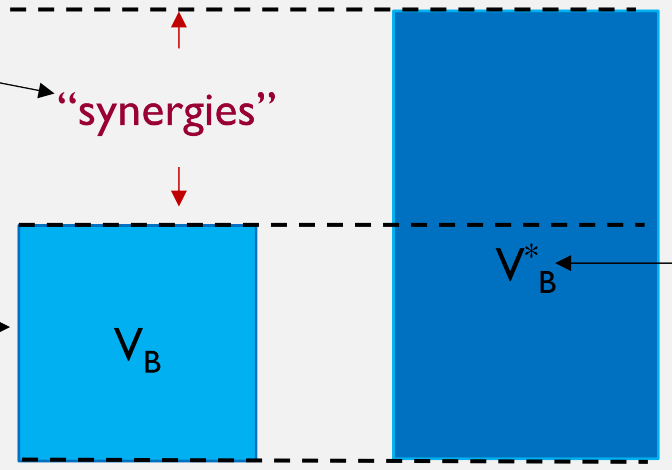


Cost of Debt Synergy Modeling
(please refer to the spreadsheet)



An alternative way to estimate synergies (& other merger gains)
Method 2: The shortcut

Step 3 Synergies $=V^*_B - V_B$



Step 1: The target's standalone total firm value (Discount rate: use $WACC_B$)

Step 2: The target's total firm value, including synergies;

- <u>Note: for synergies related to the discount rate (financial synergies):</u>
    - Use the **target's** unlevered beta.
    - Use the **acquirer's** post-merger cost of debt (r_d), tax rate (T_c), and D/E ratio.

Use the Spreadsheet " Model 5 synergy method II example blank"

- Tasks:
- 1. Estimate the value of the "cost savings synergy" or hard synergy.
- 2. Estimate the value of the "revenue enhancement synergy" or soft synergy.

Evidence on synergies in real Practice

- Evidence 1: Combined stock market reaction <u>around the announcement day</u>
  - We can examine the total stock return of the acquirer and the target on the M&A announcement date.
  - For example:
    - The acquirer's market cap is $10 billion, and the acquirer's stock return on the M&A announcement date = -1%.
    - The target's market cap is $6 billion, and the target's stock return on the M&A announcement date = 20%.
    - $Combined\ announcement\ stock\ return = \frac{10}{10+6} * (-1\%) + \frac{6}{10+6} * (20\%) = +6.88\%$
    - This +6.88% of total stock return means the stock market believes there are synergies of this M&A deal.
- Evidence from M&A history suggest that on average the *combined* announcement stock return is positive (Source: Handbook of Corporate Finance, Vol. 2). What does it mean?

Evidence on Synergies, Cont'd

- Evidence2: Acquirer's post-M&A long-run performance
  - Some prior studies show 44 to 50 percent of M&As were reversed (i.e., divested) after a long period (10 years or longer).
  - McKinsey and Co. examined 115 mergers in U.S. and U.K.
    - 60% of the M&As destroyed value for the acquirer.
    - 23% of the M&As created value for the acquirer.
  - KPMG examined 700 M&As between 1996 and 1998
    - 53% of the deals destroyed value for the acquirer.
    - 30% were neutral
    - 17% created value for the acquirer.
  - and that only a small proportion of mergers realise significant synergy for *the acquirer* after M&As.

The stock market reaction suggest that the **expected** synergies actually exist but M&As often destroy shareholders' value of the acquirer. Why?

Even if the stock market suggests that there is synergy in an M&A deal…

Overpay
The acquirer tends to pay an acquisition premium, which means most M&A benefit goes to the target.

Hidden issues
The target could have some potential issues that the acquirer and the market are not aware of.

Unsuccessful PMI
Synergy will not realize automatically. It requires a lot of work in integration.